MILITARY SPECIFICATION

ELECTRONICS ASSEMBLY, FOR MINES, M718Al/M741Al, SYSTEM

This specification is approved for use by the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements not normally covered by drawings and provides Quality Assurance provisions for the fabrication of parts, assembly and packing of the electronic assembly used in the M718El/M741El, 155MM projectile, AT/AV Mine.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-l9500	-	Semiconductor Devices General Specification For.
MIL-C-39003	-	Capacitor, Fixed, Electrolytic.
MIL-R-39008	-	Resistors, Fixed, Composition. (Insulated)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-R-39008/l - Resistors, Fixed, Composition (Insulated) Established Reliability Style RCR07.

MIL-C-39014 – Capacitor, Fixed, Ceramic.

MIL-S-45743 – Soldering, Manual Type, High Reliability, Electrical and Electronic Equipment.

MIL-S-46844 – Solder Bath Soldering of Printed Wiring Assemblies.

MIL-A-48078 – Ammunition, Standard Quality Assurance Provisions, General Specifications For.

MIL-R-55182 – Resistor, Fixed, Film.

MIL-P-63238 – Projectile, 155MM, AT M718/M741 Loading, Assembling and Packing Of.

MIL-A-63242 – Amplifier, Operation, Triple Programmable Micropower.

STANDARDS

MILITARY

MIL-STD-105 – Sampling Procedures and Tables For Inspection by Attributers.

MIL-STD-202 – Test Methods For Electronic and Electrical Component Parts.

MIL-STD-750 – Test Methods For Semiconductor Devices.

MIL-STD-883 – Test Methods and Procedures For Microelectronics.

MIL-STD-1169 – Packaging, Packing and Marking For Shipment of Inert Ammunition Components.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent herein.

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DRAWINGS

U. S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9277745	Cover, Front.
9287096	Capacitor, Fixed Electrolytic.
9287115	Potting Material.
9287126	Capacitor, Fixed Electrolytic, Tantalum, Solid.
9287185	Terminal.
9292220	Capacitor, Tantalum, Dry Electrolyte Polar, Reliability Requirements For.
9298512	Switch Assembly.
9317646	qualification Electronic Components.
9342551-1,-2	Electronic Assembly (M718).
9342552-1,-2	Electronic Assembly (M741).
9342553	Printed Wiring Board Assembly.
9342554	printed Wiring Board Assembly.
9342555	Coil Assembly.
9342558	Capacitor, Fixed Plastic Dielectric.
9342560	Resistor, Fixed Film.
9342561	Integrated Circuit, CMOS, Comparator/op Amp.
9342562	Integrated Circuit, CMOS, Quad Analog Switch.
9342564	Capacitor, Fixed Ceramic, Glass Sealed, Axial Leads.
9342574	Integrated Circuit, CMOS, Timing.
9342576	Performance Requirement (Supplemental) .
9342577	Integrated Circuit, CMOS, Logic.
9342580	Power Supply Assembly.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D2240-68 - Indentation Hardness of Rubber and Plastic by Means of a Durometer, Test For.

ASTM D412-68 - Tension Testing of Vulcanized Rubber.

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(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA., 19102).

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on drawing 9342551-1,-2, and 9342552-1,-2, and 9342576, and associated drawings and with all requirements specified in applicable specifications and standards.

3.3 Physical characteristics.

3.3.1 Envelope and dimensions. The envelope and dimensions of the electronic assembly shall conform to the requirements on drawings 9342551 or 9342552 as applicable.

3.3.2 Potting. Potting material shall comply with the requirements of drawing 9287115 and 9287148 and shall be manufactured and dispensed in accordance with the processes, procedures and controls delineated therein. The potted electronic assembly shall not show evidence of voids and/or bubbles or workmanship defects exceeding the following criteria:

a) Bubbles that are less than 1/4 inch in the maximum physical dimension are permissible. The total volume occupied by bubbles or voids cannot exceed 1/8 cu. in. (1/2 X 1/2 X 1/2" cube). In addition, no bubbles shall be in contact with any electronic component except that the power supply is allowed to trap bubbles under the lip of the battery on the contact end. (A bubble is defined to be entrapped air with no exposure to the surface.)

b) Single or multiple voids less than 1/4 inch in the maximum physical dimension are permissible on the surface except as defined in c and d. The total volume occupied by bubbles or voids cannot exceed 1/8 cu. in. (1/2 X 1/2 X 1/2" cube) and must not expose any components or the printed circuit board. (A void is defined as an irregularity of any shape which is open to the surface.)

c) Voids on the outer circumference of the lens are permissible provided no component or part of the printed wiring board is exposed.

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d) Voids on the flat vertical wall of the “D Hole” (and associated channels) up to a maximum of 25% of the depth will be allowed.

e) Sprue cutoff deviations shall be allowed to +.000” and -.040” of the curved surface.

f) Slight irregularities on the spherical surface (associated with the potting prematurely separating from the mold) within a 1" radius of the "Sprue" center are permissible.

g) A potting void or hole at the bottom of the “D” hole, exposing the battery case, of a maximum size of 1/8 inch long by 1/16 inch wide or 3/32 inch diameter is allowable provided the battery case is below bottom surface of the “D” hole. Larger voids will have potting material applied to cover the battery case and allowed to air dry.

h) Delamination of potting from the printed wiring board shall not be allowed (non-destructively viewed from the top of the board only). Components and item designation (M718E1/M741E1 shall be clearly visible through the potting material.

3.4 Environmental conditions.

3.4.1 Temperature. The electronic assembly shall perform as required herein and in drawing 9342576 over a temperature range from -25°F to +125°F.

3.4.2 Ballistic test. The electronic assembly shall perform as required after exposure to gun launch and ground impact environments of zone 8 in the M109A1 gun at 145°F. (See 4.5.25)

3.5 Electrical requirements.

3.5.1 Quiescent current. Current drawing shall be less than 300 microampere when powered by a supply voltage (VB) of 7.0 volts.

3.5.2 Self-destruct timing control and firing circuit characteristics. The electronic assembly shall provide timing control and firing signals as specified in drawing 9342576 and the following paragraphs. Unless otherwise specified VB = 7.0 volts and ambient temperature (TA) shall be 75°F.

3.5.2.1 Oscillator function. The primary time base (PTB) oscillator and the test time base (TTB) oscillator circuits shall oscillate at the periods specified in drawing 9342576 over the temperatures specified. (See 4.5.2, 4.5.3)

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3.5.2.2 Arming delay/sensor enable. The electronic assembly shall not propagate a fire signal prior to the arming time specified on drawing 9342576. The sensor circuit shall attain a state of readiness (state code 1) in the time defined in 9342576 over the temperatures specified. The initial firing capacitor charge rate shall not exceed 0.5 volts per second with a supply voltage of 7.0 volts (see 4.5.4, 4.5.5, and 4.5.6).

3.5.2.3 Low voltage detection. The electronic assembly shall provide a detonator fire signal (see para. 3.5.2.8) after the above arming delay when the supply voltage is reduced to a value between 5.9 volts (maximum) and 4.6 volts (minimum). These values shall apply at ambient temperature only. Over the range -25°F and +125°F, the limits shall be 6.1 and 4.4 volts respectively.

3.5.2.4 Self-destruct.

3.5.2.4.1 Part number 9342551. The electronic assembly, P/N 9342551, shall provide a detonator fire signal (see para. 3.5.2.8) at the self-destruct time specified in drawing 9342576 after initial application of voltage (VB). A total of 392,448 primary time base pulses shall have been generated at the period specified on drawing 9342576. (See 4.5.8).

3.5.2.4.2 Part number 9342552. The electronic assembly shall provide a detonator fire signal (see para. 3.5.2.8) at the self-destruct time specified in drawing 9342576 after initial application of voltage (VB). A total of 32000 primary time base pulses shall have been generated at the period specified on drawing 9342576. (see 4.5.9).

3.5.2.5 PTB oscillator check function. The electronic assembly shall provide a detonator fire signal (see para. 3.5.2.8), after the arming delay, when exactly two (2) periods of the test time base oscillator occur upon termination of the primary time base oscillations (see 4.5.6).

3.5.2.6 TTB oscillator crosscheck. The electronic assembly shall provide a detonator fire signal when the Test Time Base is interrupted for two successive cycles. This function shall be inhibited until elapse of the special delay period defined in drawing 9342576. (see 4.5.10)

3.5.2.7 Disturbance. The electronic assemblies that contain anti-disturbance switches shall provide a detonator fire signal (see para. 3.5.2.8), after the special delay period defined in drawing 9342576, when the assembly is jarred, tilted, rotated or otherwise tampered with (see 4.5.11).

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3.5.2.8 Detonator fire signal output energy. The electronic assembly shall provide energy to fire the detonators when required as described in the specification. The output energy shall be such that a minimum of three (3) volts shall be measured no more than forty (40) microseconds after a fire output signal is generated and a minimum of two (2) volts is measured at a minimum time of 120 microseconds after the fire output signal is generated. Voltages shall be measured at the output terminals across a 3.9 ohm + 5% resistive load (see 4.5.12).

3.5.2.9 Detonator fire signal duration from timer.

3.5.3 Target sensor functioning. The electronics assembly shall provide a fire signal (para. 3.5.3.1) when subjected to valid target signatures and not provide a fire signal when subjected to invalid signature. The specific conditions and the signal limit under which this function shall occur are specified in drawing 9342576.

3.5.3.1 Sensor output voltage. The output from the target sensor (test point 1) shall be a minimum of 2.5 volts when the assembly is subjected to a valid signature as specified in drawing 9342576.

3.5.4 Non-activated voltage. Each battery cell shall be inactivated and measured less than .005 volts when measured by a 107 ohm input impedance meter (see 4.5.14).

3.5.5 Soldering. All soldered electrical connections shall comply with MIL-S-46844, MIL-S-45743.

3.5.5.1 Solderability. Electronic and electric assembly parts that are to be soldered shall comply with the solderability requirements of MIL-S-45743 and MIL-S-46844 except integrated circuits shall be tested for solderability using the methods and acceptance criteria of MIL-STD-883 and diode and transistors shall be tested for solderability using the methods and acceptance criteria of MIL-STD-750 provided the leads are not gold plated. When gold plated leads are used, the provisions of MIL-S-45743 apply.

3.5.5.2 Quality assurance soldering plan. The contractor will generate a Quality Assurance Soldering Plan that will detail the overall soldering procedure. This plan shall include detailed information for handling, cleaning, hand soldering, machine soldering, process controls including documentation required by 3.4.4 of MIL-S-46844 tool maintenance, flagging procedures for avoiding excessive defects and all other pertinent QA procedures relative to soldering. This plan will be generated and submitted to AMCCOM, DRSMC-QAT-M(D) (see 4.5.22).

3.5.6 Shorting bar continutity. There shall be no more than one (1) ohm resistance in the shorting bar loop when measured between the designated terminals (see 4.5.16).

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3.6 Workmanship. The electronic assemblies shall be fabricated and finished in a thorough workmanlike manner. As a minimum, there shall be no potting material on the cover flange or in the battery primer hole. Sprue holes shall be ground to flush or below flush. All marking shall be neat and sharply defined. The cover shall be devoid of dirt, grease or other foreign matter. The surface coating shall be continuous except for a few light scratches not exposing base metal. All parts shall be free from chips, dirt, grease and other foreign matter.

3.7 Inspection data. Data shall be recorded and retained for all First Article and Quality Conformance tests required by this specification in accordance with 4.5.21.

3.7.1 Variables data. The primary time base oscillator period, test time base oscillator period, quiescent current, low voltage detection level, sensor threshold level shall be measured and recorded as variables data for the first article samples.

3.9 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.10 Output terminals. The output terminal well shall not contain any potting material flashing nor shall the terminal pins have any evidence of potting material adhering to the pins extending from the flat bottom section of the well. The terminal pins shall not have any evidence of exposed base metal and shall meet the solderability requirements of MIL-S-45743.

3.11 Qualification testing of electrical components. Component level qualification requirements for integrated circuits, and discrete transistors, resistors, diodes and capacitors will be as defined in 9317646.

3.12 Requalification testing of electrical components. All integrated circuits, discrete transistors, resistors, diodes and capacitors shall be subjected to another Qualification test as stipulated in the contract if:

a. The contractor’s manufacturing techniques or process which affect the form, fit, function or internal geometry of the component are revised by either the contractor or one of his sub-contractors supplying parts or assemblies.

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b. The contractor relocate his assembly facilities.

The contractor uses a new sub-contractor for material, parts, or assembly.

d. A change occurs in devices' material, performance assembly, or design.

e. The contractor ceased production for more than 90 days.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein, or in the contract, the provisions of MIL-A-48078 shall apply, and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item.

a. First article inspection

b. Quality conformance inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample, as designated by the Contracting Officer, for evaluation in accordance with the provisions of 4.3.2 and 4.3.3. The first article sample shall consist of the parts/assemblies listed below in the quantities indicated:

Part Description	Drawing	Quantity
Electronic Assembly	9342551-1 9342551-2	90 See Table 1
Electronic Assembly	9342552-1 9342552-2	90 See Table 1
Printed Wiring Board Assembly (with AD switch)	9342553	10
Printed Wiring Board	9287141	10
Coil Assembly	9342555	10

4.3.2 Inspections to be performed. See MIL-A-48078 and Table 1 specified herein.

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4.3.3 Rejection. See MIL-A-48078. The first article sample shall be rejected if any part, sub-assembly or assembly fails to comply with any of the examinations or tests specified in this specification or any of the requirements of the applicable drawings. In addition, the first article samples shall be rejected if the ballistic test sample fails as described below:

P/N 9342551-1 & -2

a. Any one (1) electronic assembly functions in any mode after expiration of the self-destruct time out (Hazardous dud).

b. Five (5) or more electronic assemblies function prior to the minimum self-destruct time or prior to being subjected to a valid target signature (early).

c. A total of three (3) or more electronic assemblies fail to function when subjected to a valid target signature.

d. A total of four (4) or more electronic assemblies fail to function in any mode (Non-hazardous dud).

P/N 9342552-1 & -2

a. Any one (1) electronic assembly functions in any mode after expiration of the self-destruct time out (Hazardous dud).

b. Five (5) or more electronic assemblies function prior to the minimum self-destruct time or prior to being subjected to a valid target signature (early).

c. A total of three (3) or more electronic assemblies fail to function when subjected to a valid target signature.

d. A total of four (4) or more electronic assemblies fail to function in any mode (Non-hazardous dud).

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-E-48630(AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
.	Electronic Assembly				See Below
					NEXT HIGHER ASSEMBLY
					--
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Electronic Assembly (Dwg. 9342551-1)				
	Examination for defects	90		3.2	4.4.2.27
	Ballistic test	Note 1		3.4.2	4.5.25
	(Dwg. 9342551-2)				
	Examination for defects	Note 1		3.2	4.4.2.27
	Ballistic test	Note 1		3.4.2	4.5.25
	Electronic Assembly (Dwg. 9342552-1)				
	Examination for defects	90		3.2	4.4.2.28
	Ballistic test	Note 2		3.4.2	4.5.25
	(Dwg. 9342552-2)				
	Examination for defects	Note 2		3.2	4.4.2.28
	Ballistic test	Note 2		3.4.2	4.5.25
	Printed Wiring Board Assy. (Dwg. 9342553)				
	Examination for defects	10		3.2	4.4.2.25/4.4.2.26

NOTES:
(1) Quantity required for five (5) rounds per Dwg. 9317597 requirements.
(2) Quantity required for five (5) rounds per Dwg. 9317598 requirements.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-48630(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
.	Electronic Assembly				See Below
					NEXT HIGHER ASSEMBLY
					--
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Coil Assembly (Dwg. 9342555)				
	Examination for defects	10		3.2	4.4.2.3
	Printed Wiring Board (Dwg. 9342554)				
	Examination for defects	10		3.2	4.4.2.22
	Solderability	10		3.2	4.4.2.22

NOTES:

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provision of MIL-A-48078. In addition, inspection lots of Electronic Assemblies and Sub-assemblies shall contain:

a. Batteries and/or power supply assemblies bearing the same lot interfix number from one manufacturer.

b. Triple operational amplifiers bearing the same lot interfix number from one manufacturer.

c. Integrated circuit, CMOS, target sensors bearing the same lot interfix number from one manufacturer.

d. Integrated circuit, CMOS, timers bearing the same lot interfix number from one manufacturer.

e. Integrated circuit, CMOS, Logic Array bearing the same lot interfix number from one manufacturer.

f. Electronic assemblies of one part number only.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-48630(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Cover, Front				9277745
					NEXT HIGHER ASSEMBLY
					9342551/9342552
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.2	Gage
102	Diameter datum A		0.40%	3.2	Gage
103	Height		0.40%	3.2	Gage
104	Spherical radius		0.40%	3.2	Comparator/Gage
105	True position of OD to datum A		0.40%	3.2	Gage
Minor					
201	Thickness		0.65%	3.2	Gage
202	Finish missing or incomplete		0.65%	3.2	Visual
203	Poor workmanship		0.65%	3.6	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-48630(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Capacitor, Fixed, Electrolytic, Tantalum, Solid				9287096
					NEXT HIGHER ASSEMBLY
					9342553
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Capacitance		0.40%	3.2	MIL-STD-202, Method 305
102	Leakage current		0.40%	3.2	MIL-STD-202, Method 302
103	Weld strength (two sides)	8	Note 1	3.2	Pull Tester
104	Internal workmanship	20	Note 2	3.2	9292220
105	Breakdown voltage	10	Note 2	3.2	9292220
106	Dissipation factor		0.40%	3.2	MIL-STD-202, Method 306
107	Solderability (2 leads)	10	Note 1	3.2	MIL-STD-202, Method 208
108	Sleeve missing	10	0.40%	3.2	Visual
Minor					
201	Poor workmanship		0.65%	3.6	Visual

NOTES: (1) Reject for two (2) or more leads/weld joints.
(2) Reject for three (3) or more defects.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-48630(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9287126
4.4.2.3	Capacitor, Fixed, Electrolytic, Tantalum, Solid				NEXT HIGHER ASSEMBLY 9342553
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Capacitance		0.40%	3.2	MIL-STD-202, Method 305
102	Leakage current		0.40%	3.2	MIL-STD-202, Method 302
103	Dissipation factor		0.40%	3.2	MIL-STD-202, Method 306
104	Weld strength (two sides)	8	Note 1	3.2	Pull Tester
105	Internal workmanship	20	Note 2	3.2	9292220
106	Breakdown voltage	10	0.40%	3.2	9292220
107	Sleeve missing		0.40%	3.2	Visual
108	Solderability (2 leads)	10	Note 1	3.2	MIL-STD-202, Method 208
Minor					
201	Poor workmanship		0.65%	3.6	Visual

NOTES: (1) Reject for two (2) or more defective.
(2) Reject for three (3) or more defective.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-48630(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Terminal				9287185
					NEXT HIGHER ASSEMBLY 9342553
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Solderability (2 leads)	20	Note 1	3.2	MIL-STD-202, Method 208
Minor 201	Poor workmanship		0.65%	3.6	Visual

NOTES: (1) Test all leads, reject for two (2) or more defective.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-48630(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	A. D. Switch Assembly				9298512
					NEXT HIGHER ASSEMBLY 9342553
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Switch function seven (7) contact closures minimum each direction		0.40%	3.2	4.5.17.1
102	Contact closure duration 0.60 seconds max.		0.40%	3.2	4.5.17.1
103	Insulation resistance		0.40%	3.2	4.5.17.2
104	Assembly damaged, distorted or incomplete		0.40%	3.2	Visual
105	Ribbon improperly welded to can		0.40%	3.2	Visual
106	Epoxy seal missing or incomplete		0.40%	3.2	Visual
107	Solderability, 2 leads			3.5.5.1	Visual
Minor					
201	Poor workmanship		1.0%	3.12	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-48630(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9342554
4.4.2.6	Printed Wiring Board				NEXT HIGHER ASSEMBLY 9342553
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Hole diameters	5	Note 1	3.2	Gage - Note 2
102	Solderability	5	Note 1	3.5.5.1	4.5.22
103	Plated thru hole integrity	5	Note 1	3.2	Visual
104	Compliance of Board to Dwg. 9342554		Lot		Certification to Note 1 of drawing
Minor					
201	Poor workmanship		0.65%	3.6	Visual

NOTES:
(1) The sample size is 32 boards. The AQL applies to the number of holes per board. If any board exceeds this criteria, the lot shall be rejected. AQL is .40%.
(2) Hole diameter undersize by .002 inch are acceptable if caused by reflow buildup.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-48630(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Coil Assembly				9342555
					NEXT HIGHER ASSEMBLY 9292216
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Winding direction (Note 2 of dwg.)	32	Note 1	3.2	Visual
102	Loose turns or broken wire	32	Note 1	3.2	Visual
103	Coil resistance 2800 Ω ± 10%	32	Note 1	3.2	Ohmmeter (Note 2)
Minor					
201	None defined		0.65%	3.6	Visual

NOTES:
(1) Reject lot if two (2) defective found.
(2) Measured at 75o ± 10oF.

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-48630(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9342558
4.4.2.8	Capacitor, Fixed Plastic Dielectric				NEXT HIGHER ASSEMBLY 9342553
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Capacitance		0.40%	3.2	MIL-STD-202, Method 30
102	Leakage current		0.40%	3.2	MIL-STD-202, Method 30
103	Dissipation factor		0.40%	3.2	MIL-STD-202, Method 30
104	Solderability (2 leads)	10	Note 1	3.2	MIL-STD-202, Method 20
Minor					
201	Poor workmanship		0.65%	3.6	Visual

NOTES: (1) Test all leads, reject for one (1) or more defective.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-48630(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.9	Semiconductor Device,Voltage Reference		9342559
			NEXT HIGHER ASSEMBLY 9342553

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Solderability (2 leads)	10	Note 1	3.2	MIL-STD-202, Method 208
102	Electrical characteristics		0.40%	3.2	Notes B thru E of dwg.
Minor					
201	Poor workmanship		0.65%	3.6	Visual

NOTES: (1) Test all leads, reject for one (1) or more defective leads.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48630(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.10	Resistor, Fixed Film				9342560-X
					NEXT HIGHER ASSEMBLY 9342553
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Resistance		0.40%	3.2	MIL-R-55182
102	Solderability (2 leads)	10	Note 1	3.2	MIL-STD-202, Method 208
Minor					
201	Poor workmanship		0.65%	3.6	Visual
202	Identification and marking		0.65%	3.2	Visual

NOTES:
(1) Test all leads, reject for one (1) or more defective leads.

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Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-48630(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Integrated Circuit, CMOS, Dual OP Amp - Dual Comparator	1 OF 1	9342561
			NEXT HIGHER ASSEMBLY 9342553

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	In Process screening		Lot		Cerfication to Note 5.2 of drawing
102	Physical dimensions	3-0-1		3.2	MIL-STD-883,Method 2016
103	Bond strength	5-0-1	Note 1	3.2	MIL-STD 883,Method 2011
104	Solderability, all leads	5-0-1	Note 2	3.2	MIL-STD-883,Method 2003
105	Physical dimension	3-0-1		3.2	MIL-STD-883,Method 2016
106	Marking	3-0-1		3.2	MIL-STD-883,Method 2015 (except 2.1.d)
107	Electrical characteristic		Lot		Certification to Note 4.4 of drawing
Minor					
201	External visual		1.0%	3.6	MIL-STD-883,Method 2009

NOTES:
(1) Pull all wires on each device, reject for one (1) or more bond failures.
(2) Test all leads, reject for one (1) or more defective leads.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-OA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-48630(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.12	Integrated Circuit, CMOS, Quad Bi Lateral Switch				9342562
					NEXT HIGHER ASSEMBLY 9342553
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	In process screening		Lot		Certification to Note 4.2 of Drawing
102	Physical dimension	3-0-1		3.2	MIL-STD-883,Method 2016
103	Bond strength, all wires	5-0-1	Note 1	3.2	MIL-STD-883,Method 2011
104	Solderability, all leads	5-0-1	Note 2	3.2	MIL-STD-883,Method 2003
105	Physical dimension	3-0-1		3.2	MIL-STD-883,Method 2016
106	Marking	3-0-1		3.2	MIL-STD-883,Method 2015 (except 2.1.d)
107	Electrical characteristic		Lot		Certification to Note 3.4 of Drawing
Minor					
201	External visual		1.0%	3.6	MIL-STD-883,Method 2009

NOTES: (1) Pull all wires on each device, reject for one (1) or more bond failures.
(2) Test all leads, reject for one (1) or more defective leads.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-48630(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.13	Capacitance, Fixed, Ceramic, Dielectric Axial Lead Glass Sealed				9342564
					NEXT HIGHER ASSEMBLY 9342553
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Capacitance		0.40%	3.2	MIL-STD-202, Method 305
102	(Leakage current) insulation resistance		0.40%	3.2	MIL-STD-202, Method 302
103	Dissipation factor		0.40%	3.2	MIL-STD-202, Method 306
104	Solderability, 2 leads	10	Note 1	3.2	MIL-STD-202, Method 208
Minor					
201	Poor workmanship		0.65%	3.6	Visual

NOTES: (1) Test all leads, reject for one (1) or more defects.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-48630(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.14	Integrated Circuit, CMOS, Timing				MIL-I-48632
					NEXT HIGHER ASSEMBLY 9342553
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	In process screening		Lot	3.2	Certification
102	Physical dimension	3-0-1		3.2	MIL-STD-883,Method 2016
103	Bond strength, all wires	5-0-1	Note 1	3.2	MIL-STD-883,Method 2011
104	Solderability, all leads	5-0-1	Note 2	3.2	MIL-STD-883,Method 2003
105	Physical dimension	3-0-1		3.2	MIL-STD-883,Method 2016
106	Marking	3-0-1		3.2	MIL-STD-883,Method 2015 (except 2.1.d)
Minor					
201	External visual		1%	3.6	MIL-STD-883,Method 2009

NOTES: (1) Pull all wires on each device reject for two (2) or more defective wires.
(2) Test all leads, reject for one (1) or more defective leads.

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Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-48630(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.15	Integrated Circuit, CMOS Logic Array	1 of 1	MIL-I-48634
			NEXT HIGHER ASSEMBLY: 9342553

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	In process screening		Lot	3.2	Certification
102	Physical dimension	3-0-1		3.2	MIL-STD-883,Method 2016
103	Bond strength, all wires	5-0-1	Note 1	3.2	MIL-STD-883,Method 2011
104	Solderability, all leads	5-0-1	Note 2	3.2	MIL-STD-883,Method 2003
105	Physical dimension	3-0-1		3.2	MIL-STD-883,Method 2016
106	Marking	3-0-1		3.2	MIL-STD-883,Method 2015 (except 2.1.d)
Minor					
201	External visual		1.0%	3.6	MIL-STD-883,Method 2009

NOTES:
(1) Pull all wires on each device, reject for two (2) or more defective wires.
(2) Test all leads, reject for one (1) or more defective leads.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-48630(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.16	Power Supply Assembly				9342580
					NEXT HIGHER ASSEMBLY 9342553
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Solderability (2 leads)	20	Note 1	3.2	MIL-STD-202, Method 208
102	Weld strength	20	Note 1	3.2	Pull Tester
103	Presence of required insulators, spacers, and tap (Note 7 of drawing)		100%	3.2	Visual
Minor					
201	Poor workmanship		0.65%	3.6	Visual

NOTES: (1) Test all leads, reject for one (1) or more defective leads.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-48630(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.17	Triple Operational Amlifier				MIL-A-63242
					NEXT HIGHER ASSEMBLY
					9342553
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Contractor shall furnisish evidence of having satisfactorily met inspection requirement of MIL-A-63242.	Lot		3.2	Certification

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-48630(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.18	Resistor				RCR07GXXX/MIL-R-39008/
					NEXT HIGHER ASSEMBLY 9342553
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Solderability (2 leads)	10	Note 1	3.2	MIL-STD-202, Method 208
102	Resistance		0.40%	3.2	MIL-R-39008/1
Minor					
201	Identification and marking (Note 2)		0.65%	3.2	Visual
202	Poor workmanship		0.65%	3.6	Visual

NOTES:
(1) Test all leads, reject for one (1) or more defective leads.
(2) Refer to drawing for resistance value.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-48630(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.19	Resistor, Fixed Film				RNC50HXXXXFS/MIL-R-55182
					NEXT HIGHER ASSEMBLY
					9342553
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Solderabiity (2 leads)	10	Note 1	3.2	MIL-STD-202, Method 208
102	Resistance		0.40%	3.2	MIL-R-55187-7
Minor					
201	Identification and marking (Note 2)		0.65%	3.2	Visual
202	Poor workmanship		0.65%	3.6	Visual

NOTES: (1) Test all leads, reject for one (1) or more defective leads.
(2) Refer to drawing for resistance value.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-48630(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.20	Semiconductor Device, Diode, Type JAN IN645-1				MIL-S-19500
					NEXT HIGHER ASSEMBLY
					9342553
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Reverse leakage current		0.40%	3.2	MIL-STD-750,Method 4016
102	Forward voltage		0.40%	3.2	MIL-STD-750,Method 4011
103	Solderability (Note 1)	10	0-1	3.5.3.2	MIL-STD-750,Method 2026
104	Evidence of JAN marking	10	0-1	3.2	Visual
Minor					
201	Poor workmanship		0.65%	3.6	Visual

NOTES: (1) Test all leads; reject for one (1) or more defective leads.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-48630(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.21	Transistor (JAN 2N222A)	1 OF 1	9287122 MIL-S-19500/255 NEXT HIGHER ASSEMBLY 9342553

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	MIL-S-19500		Lot		Certification of Compliance
102	Solderability (3 Leads)	10	Note 1	3.2	MIL-STD-750, Method 2026
103	Leakage current (ICE)		0.40%	3.2	MIL-STD-750, Method 3041 Cond. B
104	Internal visual		Lot	3.2	Certification to dwg. 9287122
105	Wire material	32	Note 1	3.2	Visual
106	Wire bond	32	Note 1	3.2	Pull tester
Minor					
201	Poor workmanship		0.65%	3.6	Visual

NOTES: (1) Test all leads/wires; reject for one (1) or more defective leads.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-48630(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.22	Transister (JAN 2N2907A)				9287122-1/MIL-S-19500/291
					NEXT HIGHER ASSEMBLY
					9342553
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	MIL-S-19500		Lot		Certification of Compliance
102	Solderability (3 leads)	10	Note 1	3.5.5.1	MIL-STD-750,Method 2026
103	Leakage current (IECE)		0.40%	3.2	MIL-STD-750,Method 3041 Cond. B
104	Internal visual		Lot	3.2	Certification to dwg. 9287122
105	Wire material	32	Note 1	3.2	Visual
106	Wire bond	32	Note 1	3.2	Pull tester
Minor					
201	Poor workmanship		0.65%	3.6	Visual

NOTES: (1) Test all leads/wires, reject for one (1) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-48630(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.23	Printed Wire Board Assembly (Configured for 9342551)				9342553
					NEXT HIGHER ASSEMBLY
					9342551-1 & -2
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Solder workmanship		0.40%	3.5.5.1	Visual/4.5.22
102	Conformal coating missing from noted components		0.40%	3.2	Visual
103	Any component damaged, missing or improperly assembled		0.40%	3.2	Visual
104	Battery lead weld(s) broken		Note 1	3.2	Visual (5X)
Minor					
201	Height from board to top of battery mold		0.65%	3.2	Gage
202	Height of leads/solder on bottom of board		0.65%	3.2	Gage
203	Poor workmanship		0.65%	3.6	Visual

NOTES: (1) MIL-STD-105D, Level II, Reject for one (1) or more defective lead welds.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-48630(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.24	Printed Wire Board Assembly (Configured for 9342552)				9342553
					NEXT HIGHER ASSEMBLY
					9342551-1 & -2
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Solder workmanship		0.40%	3.5.5.1	Visual/4.5.22
102	Conformal coating missing from noted components		0.40%	3.2	Visual
103	Any component damaged, missing or improperly assembled		0.40%	3.2	Visual
104	Battery lead weld(s) broken		Note 1	3.2	Visual (5X)
Minor					
201	Height from board to top of battery mold		0.65%	3.2	Gage
202	Height of leads/solder on bottom of board		0.65%	3.2	Gage
203	Poor workmanship		0.65%	3.6	Visual

NOTES: (1) MIL-STD-105D, Level II, Reject for one (1) or more defective lead welds.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-48630(AR)

PARAGRAPH	TITLE		SHEET 1 OF 4		DRAWING NUMBER
4.4.2.25	Electronic Assembly				9342551-1 & -2
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	(Note 1)				
1	SD timeout high		100%	3.5.2.4.1	4.5.8
2	SD timeout high - real time test				
	Hot temperature (Note 4)		8-0-1	3.5.2.4.1	4.5.26
	Cold temperature (note 5)		8-0-1	3.5.2.4.1	4.5.26
	Ambient temperature		16-0-1	3.5.2.4.1	4.5.26
3	Max PTB oscillator period		100%	3.5.2.1	4.5.2
4	Correlation between SD jumper position and part number		100%	3.2	Visual
Special	(Note 2)				
a	Max TTB oscillator period		100%	3.5.2.1	4.5.3
b	PTB Oscillator check circuit function		100%	3.5.2.5	4.5.6
c	TTB Oscillator cross check		100%	3.5.2.6	4.5.10
d	Low voltage circuit trigger level detect function		100%	3.5.2.3	4.5.7
e	Self-sterilization circuit function		100%	3.5.2.8	4.5.12
NOTE (1)	Report all critical defects per 4.5.21.				
NOTE (2)	See para 6.7.				
NOTE (3)	16-0-2 Dud				
	32-1-2 Dud (Double Sample)				
NOTES:	16-1-3 Early				
	32-2-3 Early (Double Sample)				

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-48630(AR)

PARAGRAPH	TITLE			SHEET 2 OF 4	DRAWING NUMBER 9342551-1 & -2
4.4.2.25 Cont'd	Electronic Assembly				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101	Min PTB oscillator period	4.4.3.1		3.5.2.1	4.5.2
102	Min TTB oscillator period	4.4.3.1		3.5.2.1	4.5.3
103	Arming delay/sensor enable	4.4.3.1		3.5.2.2	4.5.4/4.5.5/4.5.6
104	SD timeout low	4.4.3.1		3.5.2.2 4.1	4.5.7
105	AD switch mode (9342551-2, Ambient only)	Note 6		3.5.2.7	4.5.11
106	Shorting bar continuity	4.4.3.1		3.5.6	4.5.16
107	Output energy	4.4.3.1		3.5.2.8	4.5.12
108	Target sensor function	4.4.3.1		3.5.2.8	4.5.13
109	Quiescent current	4.4.3.1		3.5.1	4.5.1
110	Potting material characteristics	Note 7		3.3.2	4.5.15
111	Excessive voids/bubbles in potting		0.40%	3.3.2	Visual/Gage
112	Spherical radius profile		0.40%	3.2	Gage
113	Depth of "D" hole		0.40%	3.2	Gage
114	Diameter of "D" hole		0.40%	3.2	Gage
115	True position of "D" hole to center hole		0.40%	3.2	Gage

NOTE (6)

Lot Size	Sample
0-500	12-1-2
501-3200	45-2-3
3201-10,000	68-3-4
10,001-35,000	105-4-5

NOTE (7) Sample in accordance with 4.4.3.2. Potting process shall be verified daily.

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-48630(AR)

PARAGRAPH	TITLE		SHEET 3 OF 4		DRAWING NUMBER
4.4.2.25 Cont'd	Electronic Assembly				9342551-1 & -2
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
116	True position of OD to center hole		0.40%	3.2	Gage
117	Height from flange to top of "D" hole		0.40%	3.2	Gage
118	Depth from the bottom of "D" hole to the lower shoulder of the mid-section center hole		0.40%	3.2	Gage
119	ID (min) of the mid section center hole		0.40%	3.2	Gage
120	ID (min) of the top section center hole		0.40%	3.2	Gage
121	Terminal(s) extending above spherical surface		0.40%	3.2	Visual
122	Presence of AD switch (9342551-2 only)		0.40%	3.2	Visual
123	Real time SD timeout low - Dud and early				4.5.26
	Hot temperature (Note 4)	8-1-3 16-2-3		3.5.2. 4.1	4.5.26
	Cold temperature (Note 5)	8-1-3 16-2-3		3.5.2. 4.1	4.5.26
	Ambient (Note 3)	16		3.5.2. 4.1	4.5.26
124	One or both battery cells activated		100%	3.5.4	4.5.14
125	Output terminal(s) with excess potting material or exposed base	4.4.3.3		3.10	Visual, 4X Magnifying Lamp

NOTES:
(4) Same unit shall be used for Major 123, hot temperature tests.
(5) Same units shall be used for Major 123 cold temperature tests.

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-48630(AR)

PARAGRAPH	TITLE		SHEET 4 OF 4		DRAWING NUMBER
4.4.2.25 Cont'd	Electronic Assembly				9342551-1 & -2
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Width of test pad hole		0.65%	3.2	Gage
202	Length of test pad hole		0.65%	3.2	Gage
203	Poor workmanship		0.65%	3.6	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-48630(AR)

PARAGRAPH	TITLE		SHEET 1 OF 4		DRAWING NUMBER
4.4.2.26	Electronic Assembly				934255-2-1 & 2
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	(Note 1)				
1	SD timeout high		100%	3.5.2.4.1	4.5.9
2	SD timeout high - real time test				
	Hot temperature (Note 4)		8-0-1	3.5.2.4.1	4.5.26
	Cold temperature (Note 5)		8-0-1	3.5.2.4.1	4.5.26
	Ambient temperature		16-0-1	3.5.2.4.1	4.5.26
3	Max PTB oscillator period		100%	3.5.2.1	4.5.2
4	Correlation between SD jumper position and part number		100%	3.2	Visual
Special	(Note 2)				
a	Max TTB oscillator period		100%	3.5.2.1	4.5.3
b	PTB Oscillator check circuit function		100%	3.5.2.6	4.5.6
c	TTB Oscillator cross check		100%	3.5.2.6	4.5.10
d	Low voltage circuit trigger level detect function		100%	3.5.2.3	4.5.7
e	Self-sterilization circuit function		100%	3.5.2.8	4.5.12
NOTE (1)	Report all critical defects per 4.5.21				
NOTE (2)	See para 6.7				
NOTE (3)	16-0-2 Dud				
NOTES:	32-1-2 Dud (Double Sample)				
	16-1-3 Early				
	32-2-3 Early (Double Sample)				

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-48630(AR)

PARAGRAPH	TITLE		SHEET 2 OF 4		DRAWING NUMBER
4.4.2.26 Cont'd	Electronic Assembly				9342552-1 & -2
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101	Min PTB oscillator period	4.4.3.1		3.5.2.1	4.5.2
102	Min TTB oscillator period	4.4.3.1		3.5.2.1	4.5.3
103	Arming delay/sensor enable	4.4.3.1		3.5.2.2	4.5.4/4.5.5/4.5.6
104	SD timeout low	4.4.3.1		3.5.2.2	4.5.7
				4.1	
105	AD switch mode (9342552-2, Ambient only)	Note 6		3.5.2.7	4.5.11
106	Shorting bar continuity	4.4.3.1		3.5.6	4.5.16
107	Output energy	4.4.3.1		3.5.2.8	4.5.12
108	Target sensor function	4.4.3.1		3.5.2.8	4.5.13
109	Quiescent current	4.4.3.1		3.5.1	4.5.1
110	Potting material characteristics	Note 7		3.3.2	4.5.15
111	Excessive voids/bubbles in potting		0.40%	3.3.2	Visual/Gage
112	Spherical radius profile		0.40%	3.2	Gage
113	Depth of "D" hole		0.40%	3.2	Gage
114	Diameter of "D" hole		0.40%	3.2	Gage
115	True position of "D" hole to center hole		0.40%	3.2	Gage
NOTE (6)	Lot Size / Sample: 0-500 / 12-1-2; 501-3200 / 45-2-3; 3201-10,,000 / 68-3-4; 10,001-35,000 / 105-4-5				

NOTES:

(7) Sample in accordance with 4.4.3.2. Potting process shall be verified daily.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-48630(AR)

PARAGRAPH	TITLE		SHEET 3 OF 4		DRAWING NUMBER
4.4.2.26 Cont'd	Electronic Assembly				9342552-1 & -2
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
116	True position of OD to center hole		0.40%	3.2	Gage
117	Height from flange to top of "D" hole		0.40%	3.2	Gage
118	Depth from the bottom of "D" hole to the lower shoulder of the mid-section center hole		0.40%	3.2	Gage
119	ID (min) of the mid section center hole		0.40%	3.2	Gage
120	ID (min) of the top section center hole		0.40%	3.2	Gage
121	Terminal(s) extending above spherical surface		0.40%	3.2	Visual
122	Presence of AD switch (9342551-2 only)		0.40%	3.2	Visual
123	Real time SD timeout low - Dud and early				4.5.26
	Hot temperature (Note 4)	8-1-3		3.5.2 4.1	4.5.26
	Cold temperature (Note 5)	8-1-3		3.5.2 4.1	4.5.26
	Ambient (Note 3)	16		3.5.2 4.1	4.5.26
124	One or both battery cells activated		100%	3.5.4	4.5.14
125	Output terminal(s) with excess potting material or exposed base	4-4-3	3	3.10	Visual, 4X Magnifying Lamp

NOTES: (4) Same unit shall be used for Major 123 hot temperature tests.
(5) Same units shall be used for Major 123 cold temperature tests.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-48630(AR)

PARAGRAPH	TITLE		SHEET 4 OF 4		DRAWING NUMBER
4.4.2.26 Cont'd	Electronic Assembly				9342551-1 & -2
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Width of test pad hole		0.65%	3.2	Gage
202	Length of test pad hole		0.65%	3.2	Gage
203	Poor workmanship		0.65%	3.6	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection Tables and below.

4.4.3.1 Ambient (75°F + 10°F), Hot (145°F + 5°F) and Cold (-25°F -5°F) temperature functioning tests. The table below shall be used for the selection of the lot acceptance sample sizes for ambient, hot and cold temperature testing of lots of electronic assemblies.

INSPECTION LOT SIZE	SAMPLE SIZE	* AMB	HOT	COLD	ACCEPT - REJECT CRITICAL/SPECIAL DEFECTS	MAJOR DEFECTS
0-150	20	8	6	6	0-1	0-1
151-500	32	12	10	10	0-1	0-1
501-3200	125	45	40	40	0-1	1-2
3201-10,000	200	68	66	66	0-1	2-3
10,000-35,000	315	105	105	105	0-1	3-4

The units shall be tested for all the functional parameters (under critical, special, major) listed in 4.4.2.27, 4.4.2.28. Reject number is the number of defective units that would reject the lot. Since all the units have to be 100% inspected at ambient for critical and special defects, the quantities list under * do not have to be reinspected for critical and special defects at ambient temperature (see 4.5.18).

4.4.3.2 Potting material testing. The potting material shall be checked for tensile strength, and hardness in accordance with drawing 9287148. Two test samples shall be molded per the drawing for each batch of mixed material or each 1,000 electronic assemblies assembled during continuous production, or each days production, whichever comes first. One test sample will be used for testing and one retained per the drawing. Identification shall be maintained between the sample and assemblies. Failure of the test specimen to comply with requirements shall result in rejection of the lot of electronic assemblies assembled from the batch of potting material (see 4.5.15).

4.4.3.3 Inspection criteria. After three (3) consecutive lots have successfully passed 100% inspection, the contractor may switch to sample inspection at an AQL .25% in accordance with MIL-STD-105D. The above one hundred percent (100%) inspection shall be reinstated if test sample fails to comply with the above acceptance criterial level.

4.4.4 Inspection equipment. The inspection equipment required to perform the examination and tests prescribed herein is identified in the 'Paragraph Reference/Inspection Method' column in Classification of Defects and Tests forms beginning with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and section 6.3 herein.

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4.4.5 Dimensional control of molded parts. In place of normal sampling and inspections associated with the Classification of Defects and after a curing time for the parts has been approved (see note 1), the contractor may, with the approval of the Government, select a sample of three (3) parts (as molded) from each cavity to be fully inspected dimensionally to qualify a new or reworked cavity for use in production. The molded parts shall carry the individual cavity identification. As a control of each cavity during production, the above quantity of parts from each cavity shall be inspected for at least the defects listed in pertinent paragraphs 4.4.2.1 through 4.4.2.28 after continuous production of each 5,000 parts or at the end of each week, whichever occurs first, Of the three (3) samples, one (1) sample shall be the last part produced. If any defective parts are found during qualification of the cavity, the cavity producing the defective part will not be used in production. If any defective parts are found when inspection is performed for the control of the cavity, the cavity producing the defective part shall be removed from production. Further., that portion of production since the last control check shall be returned to the contractor for inspection for each separate type of defect according to MIL-STD-105, using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. All cavities removed from production because of some fault may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request. Contractor design of gages and test equipment required to perform the inspections listed herein shall be forwarded to Commander, ARDC, Dover, New Jersey 07801, ATTN: AMSMC-QAT-M(D) for approval prior to manufacture of equipment. The note subparagraphs identify this paragraph.

NOTE 1: In establishing a curing time, dimensionally check ten (10) parts from each cavity at periodic intervals (e.g. every 30, 60, etc. mins.) until dimensional stability is attained. The curing time will be from the time the part comes out of the mold until dimensional stability is attained. The inspection data used in determining the curing time shall be sent to ARDC, Dover, New Jersey 07801, ATTN: AMSMC-QAT-M(D). When a new cavity is substituted, an old cavity reworked, or the material or cycle time hanged, a new curing time shall be established and approved (see 6.4).

NOTE 2: Those features of a part which are subsequently machined or otherwise worked upon to meet its drawing requirements shall be inspected in accordance with the classification of defects.

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4.5 Test methods and procedures. Only Government approved test equipment and test fixtures shall be connected to electronics assemblies (electronic assembly) under test to verify that it complies with the requirements specified herein. Increasing the PTB oscillator frequency to reduce test time is acceptable, provided the higher frequency used does not cause or mask malfunctions in the functional operation of the electronic assembly; the frequency used is within the capability of the test equipment to monitor response with the required accuracy; and the actual procedure used to increase the frequency of the signal is approved by the Government before it is implemented. When the PTB oscillator frequency is increased, the TTB oscillator frequency is to be proportionately increased or the TTB oscillator inhibited as applicable for the test being implemented. All testing shall be performed, unless otherwise specified, at a supply voltage of 7.0 ± 0.15 volts, an ambient temperature of 75°F ± 10°F, and a relative humidity of 60 ± 20%. Also, the rise time of the supply voltage applied to the electronic assembly under test shall not be less than 5 milliseconds nor greater than 200 milliseconds. All tests are nondestructive unless otherwise noted.

4.5.1 Quiescent current test. With the electronic assembly under test connected to the test fixture and with the detonator terminal E_1 and E_2 connected through a 3.9 ± 5% ohm resistor, measure the quiescent current of the electronic assembly. The measured current shall be observed for compliance with the requirement specified in 3.5.1. This is a non-destructive test.

4.5.2 Primary time base (PTB) period test. With the electronic assembly under test connected to the test fixture and with both the primary and test time base circuits oscillating at their natural frequency, measure the period of the PTB oscillator at test point T.P.4. Observe the measured period for compliance with the requirements specified in 3.5.2.1 for the corresponding temperatures and voltages under which this test is conducted. This is a non-destructive test.

4.5.3 Test time base (TTB) period test. With the electronic assembly under test connected to the test fixture and with both the primary and test time base oscillators operating at their natural frequency, measure the period of the TTB oscillator at test point T.P.5. Observe the measured period for compliance with the requirements specified in 3.5.2.1 for the corresponding temperatures and voltages under which this test is conducted. This is a non-destructive test.

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4.5.4 Firing capacitor charge rate. Verify that the charge rate does not exceed the value stated in 3.5.2.2. This test must be done at the time of initial power application to the unit under test.

4.5.5 Sensor enable test. With the electronic assembly under test connected to the test fixture, verify that State Code 1 occurs at the time specified in 3.5.2.2. A shielded magnetic environment is required for this test. State code test points and voltage are defined in drawing 9342576.

4.5.6 PTB oscillator crosscheck and arming time check. With the electronic assembly under test connected to the test fixture and both the PTB and TTB oscillator circuits operating at their natural frequencies, verify that the oscillator check circuit is operational by stopping PTB at the maximum arming time (-0, +2 sec) specified in 3.5.2.2, and observing that the oscillator check circuit initiates the turn on of the detonator firing circuit within 2 TTB periods after the PTB oscillator is stopped. The observed fire pulse will be of reduced amplitude and the criteria of 3.5.2.8 do not apply.

4.5.7 Low voltage detect test. With the electronic assembly under test connected to the test fixture, verify that the low voltage detect circuit initiates the turn on of the detonator firing circuit at the temperature and voltage specified in 3.5.2.3. For this test, it is required that both the PTB and TTB oscillators be operating at their natural frequency, and that the initial supply voltage be 7.0 ± .15 volts, linearly reduced at a rate of 7.0 (max.) volts/second until the low voltage detect circuit is triggered.

4.5.8 Self-destruct signal test (Dwg. 9342551). With the electronic assembly under test connected to the test fixture, verify that the self-destruct signal, that initiates the turn on of the detonator firing circuit, occurs at the PTB count specified in 3.5.2.4.1.

4.5.9 Self-destruct signal test (Dwg. 9342552). With the electronic assembly under test connected to the test fixture, verify that the self-destruct signal, that initiates the turn on of the detonator firing circuit, occurs at the PTB count specified in 3.5.2.4.2.

4.5.10 TTB oscillator crosscheck. With the electronic assembly under test connected to the test fixture, short TTB to supply negative and verify that the detonator firing circuit is activated within the time period specified in 3.5.2.6.

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4.5.11 Anti-disturbance function mode. After expiration of the special delay period (dwg. 9342576), with the electronic assembly under test connected to the test fixture so that its plane makes an approximate 45 degree angle with the horizontal, the electronic lens assembly shall be moved thru an arc of 90 degrees maximum about the cylindrical axis. Observation shall be made that the detonator firing circuit turns on before the 90 degree rotation is complete. Observation shall be made for compliance with the requirements of 3.5.2. This is a non-destructive test. All defects shall be rerun at the end of the lot acceptance test, once, to verify the defect. If a unit functions properly upon retest, it shall be classified as an acceptable unit.

4.5.12 Firing circuit energy. With the electronic assembly under test connected to the test fixture, verify that the parallel detonators firing circuits meet the requirements specified in 3.5.2.7. This is a non-destructive test. This test may be combined with that of 4.5.7.

4.5.13 Magnetic target sensor functioning test. The potted electronic assembly shall be connected to approved test equipment and tested for proper functioning of the magnetic sensor circuitry. The electronic assembly shall be tested for the all-fire and no-fire levels and firing angle specified on the drawing 9342576. Countermeasure response and spurious outputs shall also be checked at this assembly level. Observation shall be made for compliance with the applicable requirements of 3.5.3. Non-destructive test.

4.5.14 Non-activated voltage. Each battery cell shall be checked for conformance with the requirements of 3.5.4 when measured by a 10 megohm input impedance volt-meter. Measurement shall be made of each cell individually in the potted electronic assembly. Non-destructive test.

4.5.15 Potting material testing. Tests of the potting material shall be done in accordance with drawing 9287148 and 4.4.3.2 herein. Observation shall be made for compliance with requirements of 3.3.2 and 4.4.3.2.

4.5.16 Shorting bar resistance. The continuity - resistance of each shorting bar circuit shall be measured, using an approved ohmmeter, between the test points E3 and TP9. Observation shall be made for compliance with requirement of 3.5.6. This is a non-destructive test. Acceptable parts may be returned to the lot.

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4.5.17 Switch assembly (Anti-disturbance switch).

4.5.17.1 AD switch function and closure time duration. The switch assembly shall be placed in an approved fixture. The fixture shall be capable of rotating the switch assembly through 360° on its cylindrical axis, while this axis is held at 15 ± 2° referenced to the horizontal, for both CW and CCW directions of rotation. Verify that the functional requirements of the drawing are met. This is a non-destructive test. Acceptable parts may be returned to the lot.

4.5.17.2 Insulation resistance of AD switch. The switch assembly when in a suitable fixture shall be measured for insulation resistance in the reset position using an approved megohmmeter. Verify that the switch assembly does meet the requirements of the drawing. This is a non-destructive test. Acceptable parts may be returned to the lot.

4.5.17.3 Plating of AD switch can. Five (5) cans shall be randomly selected from each plating batch. The specimens shall be cross-sectioned through the axial center of the can, polished and examined microscopically to measure the plating thickness. If the total average plating thickness for the five cans fails to meet the requirements as specified on the drawing, the sample shall be classed defective and the lot shall be rejected. (Destructive test).

4.5.18 Ambient, hot and old temperature functional testing. The sample of electronic assemblies specified in para. 4.4.3.1 shall be subjected to all functional tests at the operating temperature extremes. one-third of the sample specified shall be functionally tested atambient (75°F ± 10°F) and one-third at-25°F (+0°, -3°) and one-third at l45°F (+3°, -0°), (see 4.4.3.1). The units shall be conditioned at the above temperatures for two (2) hours prior to the test. Testing shall be performed with the unit in the temperature chamber. Approved functional test equipment shall be used. Acceptable threshold levels are specified on drawing 9342576. Acceptance limits for the digital circuitry (SD timing control) at the temperature extremes are specified on drawing 9342576.

4.5.19 Printed wiring board cleaning procedures. Printed wiring board assembly cleaning procedures shall be documented and supplied to the Government (see 6.6). Verification of adherence to the cleansing procedure shall be done on a daily basis (Ref. Data Item DI-T-5372 (Tailored).

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4.5.20 Integrated circuit handling procedures. CMOS integrated circuits require care in handling to avoid damage due to static discharge. The contractor shall document his handling procedure for the CMOS integrated circuits and supply the procedure to the Government (see 6.6). Adherence to the handling procedure shall be verified daily. (Ref. DI-T-5372 (Tailored)).

4.5.21 Data requirements. The contractor shall submit a quality inspection test, demonstration and evaluation report in accordance with Data Item Description DI-R-1724. It shall include all detailed inspection and test results. Test data reports shall be submitted to the technical agency listed in 6.4.

4.5.22 Quality assurance soldering plan. The QA soldering plan as noted in 3.5.5.2 shall be submitted for Government approval (see 6.6). (Ref. DI-T-5372 (Tailored)).

4.5.23 Variables data. The contractor shall submit variables data in accordance with Data Item Description DI-R-1724 for the First Article samples and surveillance units. It shall include all detailed inspection and test results with respect to the following parameters: Primary Time Base Oscillator Period, Test Time Base Oscillator Period, Quiescent Current, Low Voltage Detection Level, Magnetic Sensor Threshold Level. Test data reports shall be submitted to the technical agency listed in 6.4.

4.5.24

4.5.25 Ballistic test. Electronic assemblies shall be capable of performance as stated herein after being subjected to the ballistic environment of the 155MM M109A1 Howitzer, zone 8 and conditioned at 145°F. The electronic assemblies shall be forwarded to the load plant specified in the contract for assembly into non-hazardous test projectiles. Electronic assemblies, P/N 9342551-1 & -2, shall be loaded into Projectile, 155MM test, Non-hazardous, M718 type, Dwg. 9317597. Electronic assemblies, P/N 9342552-1 & -2, shall be loaded into Projectile, 155MM test, Non-hazardous, M741 type, Dwg. 9317598. Testing of the projectile and mines shall be in accordance with MIL-P-63238, para. 4.5.8 as applicable for the Non-hazardous type mines. Of the forty-five (45) mines of each type to be tested, nine (9) shall be randomly chosen and allowed to self-destruct per MIL-P-63238, para. 4.5.8.3.b. (1}. The remaining thirty-six (36) shall be tested for sensor performance per MIL-P-63238, para. 4.5.8.3.b. (2).

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4.5.26 Self-Destruct (SD) mode destructive tests (Real Time). A randomly selected sample of electronics assemblies P/N 9342552-1, -2/9342552-1, -2, with simulated resistive load for the parallel detonators, shall have their batteries activated by electric battery primer, drawing 9292624 and allowed to function in “Real Time” as follows:

a. Functioning shall be at hot, cold, and ambient.

b. SD functioning shall be part of lot acceptance tests. Observation shall be made for compliance with requirement of Dwg. 9287153. Units under test shall be conditioned at +145°F, -25°F, and +75°F.

c. Through use of fixtures and/or inclosures, the sample units shall be protected from radio frequency and magnetic field effect during the time interval covering activation of batteries through functioning.

d. If an electronic assembly fails to function by the maximum set self-destruct time, a valid target signature shall be applied (using a simulated prime target sensor signature coil). Any assembly failing to function (Non-hazardous dud) or functioning by sensor after SD time (Hazardous dud) shall be set aside for dismantling and failure analysis.

e. Event time shall be monitored and recorded.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The electronic assemblies shall be preserved and packaged in accordance with MIL-STD-1169, Level C.

5.2 Packing. Packing of the assemblies shall be in accordance with MIL-STD-1169, Level B.

5.3 Marking. Marking shall be in accordance with MIL-STD-1169.

5.4 Shipping. When shipments from more than one lot are shipped as a carload, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of lots in transit.

6. NOTES

6.1 Intended use. The component covered by this specification is intended for use in the M718/M741 System.

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6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number, and date of this specification.

b. Provisions for submission of first article samples.

c. Provisions of MIL-A-48078.

d. Certification of conformance.

e. Provisions for surveillance units.

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, AMCCOM, ATTN: AMSMC-QAT-M(D), Dover, New Jersey, 07801-5001.

6.4 Submission of results of contractor-conducted examinations and tests. Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations of tests to Commander, AMCCOM, ATTN: AMSMC-QAT-M(D), Dover, New Jersey, 07801-5001.

6.5 Distribution of ammunition data cards. Distribution of data cards shall include the following: Commander, AMCCOM, ATTN: AMSMC-QAT-M(D), Dover, New Jersey 07801-5001.

6.6 Submission of procedures. Submission of procedures specified in 4.4.3.2, 4.5.11, 4.5.20 and 4.5.22 shall be to Commander, AMCCOM, ATTN: AMSMC-QAT-M(D), Dover, New Jersey 07801-5001.

6.7 Special defect. A defect, other than critical that judgement and experience indicate may, depending upon the degree of variance from the design requirement.

a. Result in hazardous or unsafe conditions for individuals using, maintaining or depending upon the product or

b. Prevent performance of the tactical function of a major end item.

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6.8 Subject term (key word) listing.

Anti-disturbance function
Ballistic testing, performance survivability
Coil Asembly
Electronic Assembly
Low voltage detection function
Magnetic target sensor function
Magnetic target sensor function
Mine, Anti-tank
Potting Material
Primary time base oscillator period test
Quiescent current test
Self-Destruct time function
Test time base oscillator period test.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1345-A338)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-E-48630(AR)	ELECTRONICS ASSEMBLY, FOR MINES, M718A1/M741A1, SYSTEMS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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